Measurement Error: Models, Methods and Applications

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DEDICATION

In memory of my parents, Eugene and Jeanne, and to my wife, Elaine, and my children, Jessie and Gene.

Contents

Preface

This book is about measurement error, which includes misclassification. This occurs when some variables in a statistical model of interest cannot be observed exactly, usually due to instrument or sampling error. It describes the impacts of these errors on "naive" analyses that ignore them and presents ways to correct for them across a variety of statistical models, ranging from the simple (one-sample problems) to the more complex (some mixed and time series models) with a heavy focus on regression models in between.

The consequences of ignoring measurement error, many of which have been known for some time, can range from the nonexistent to the rather dramatic. Throughout the book attention is given to the effects of measurement error on analyses that ignore it. This is mainly because the majority of researchers do not account for measurement error, even if they are aware of its presence and potential impact. In part, this is due to a historic lack of software, but also because the information or extra data needed to correct for measurement error may not be available. Results on the bias of naive estimators often provide the added bonus of suggesting a correction method.

The more dominant thread of the book is a description of methods to correct for measurement error. Some of these methods have been in use for a while, while others are fairly new. Both misclassification and the so-called "errors-in-variables" problem in regression have a fairly long history, spanning statistical and other (e.g., econometrics and epidemiology) literature. Over the last 20 years comprehensive strategies for treating measurement error in more complex models and accounting for the use of extra data to estimate measurement error parameters have emerged. This book provides an overview of some of the main techniques and illustrates their application across a variety of models. Correction methods based on the use of known measurement error parameters, replication, internal or external validation data, or (in the case of linear models) instrumental variables are described. The emphasis in the book is on the use of some relatively simple methods: moment corrections, regression calibration, SIMEX, and modified estimating equation methods. Likelihood techniques are described in general, but only implemented in some of the examples.

There are a number of excellent books on measurement error, including

Fuller's seminal 1987 book, mainly focused on linear regression models, and the second edition of Carroll et al. (2006). The latter provides a comprehensive treatment of many topics, some of which overlap with the coverage in this book. Other, more specialized books include Cheng and Van Ness (1999) and Gustafson (2004). The latter provides Bayesian approaches for dealing with measurement error and misclassification in numerous settings. This book (i.e., the one in your hands), which is all non-Bayesian, differs from these others both in the choice of topics and in being more applied. The goal is to provide descriptions of the basic models and methods, and associated terminology, and illustrate their use. The hope is that this will be a book that is accessible to a broad audience, including applied statisticians and researchers from other disciplines with prior exposure to statistical techniques, including regression analysis. There are a few places (e.g., the last two chapters and parts of Chapters 5 and 6) where the presentation is a bit more advanced out of necessity.

Except for Chapter 6, the book is structured around the model for the true values. In brief, the strategy is to first discuss some simpler problems including misclassification in estimating a proportion and in two-way tables, as well as additive measurement error in simple and multiple regression. We then move to a broad treatment of measurement error in Chapter 6, and return to specific models for true values in the later chapters. A more detailed overview of the chapters and the organization of the book is given in Section 1.5. The layout is designed to allow a reader to easily locate a specific problem of interest. A limited number of mathematical developments, based on relatively basic theory, are isolated into separate sections for interested readers. However, I have made no attempt to duplicate the excellent theoretical coverage of certain topics in the aforementioned texts.

The measurement error literature has been growing at a tremendous rate recently. I have read a large number of other papers that have influenced my thinking but do not make their way explicitly into this book. I have tried to be careful to provide a direct reference whenever I use a result from another source. Still, given the more applied focus of the book, the desire to limit the coverage in certain areas, and size restrictions, means that many important papers are left unrecognized. I apologize beforehand to anyone who may feel slighted in any way by my omission of their work. In addition, while some very recent work is referenced in certain places, since parts of the book were completed up to two years before publication date, I have not tried to incorporate recent developments associated with certain topics. I also recognize that in the later stages of the book the coverage is based heavily on my own work (with colleagues). This is especially true in the last two chapters of the book. This is largely a matter of writing about what I know best, but it also happens to coincide with coverage of some of the more basic settings within some broader topics. When I set out, I had planned to provide more discussion about the connections between measurement error models and structural equation and latent variable models, as well as related issues with missing data and omitted covariates. There are a few side comments on this, but a fuller discussion was omitted for reasons of both time and space.

In a way this book began in the spring of 1989, when I was on my first sabbatical. In the years right before that I had wandered into the "errors-in-variables", or measurement error, world through an interest in calibration and ratios. Being a bit frustrated by the disconnect between much of the traditional statistics literature and that from other areas, especially econometrics, I began to build a systematic review and a bibliography. Fortuitously, around this time Jim Ware invited me to attend a conference held at NIH in the fall of 1989 on measurement error problems in epidemiology. I also had the opportunity to attend the IMS conference on measurement error at Humboldt State University the following summer and listen to many of the leading experts in the field. This period marked the beginning of an explosion of new work on measurement error that continues to this day. I am extremely grateful to Jim for the invitation to the NIH conference, getting me involved with the measurement error working group at Harvard and supporting some of my early work in this area through an EPA-SIMS (Environmental Protection Agency and the Societal Institute of the Mathematical Sciences) grant. I benefited greatly from interactions with some of the core members of the Harvard working group including Ernst Linder, Bernie Rosner, Donna Spiegelman and Tor Tosteson. This led to an extensive and fruitful collaboration with Tor and Eugene Demidenko (and many pleasant rides up the Connecticut River Valley to Dartmouth to work with them, and sometimes Jeff Buzas from the University of Vermont).

By the early 90s I had a preliminary set of notes and programs, portions of which were used over a number of years as part of a topics in regression course at the University of Massachusetts. The broader set became the basis for a short course taught for the Nordic Region of the Biometrics Society in Ås, Norway in 1998. These were updated and expanded for a short course taught in 2005 at the University of Oslo Medical School, and then a semester long course at the University of Massachusetts in the fall of 2007. I appreciate the feedback from the students in those various courses. The material from those courses made up the core of about half of the book. I am extremely grateful to Petter Laake for helping to arrange support for my many visits to the Medical School at the University of Olso. I have enjoyed my collaborations with him, Magne Thoresen, Marit Veirod and Ingvild Dalen and greatly appreciate the hospitality shown by them and the others in the biostatistics department there.

I'm also grateful to Ray Carroll for sharing some of the Framingham Heart Study data with me, Walt Willet for generously allowing me access to the the external validation data from the Nurses Health Study, Rob Greenberg for use of the beta-carotene data, Sandy Liebhold for the egg mass defoliation data, Joe

Elkinton for the mice density data, Paul Godfrey and the Acid Rain monitoring project for the pH and alkalinity data, and Matthew DiFranco for the urinary neopterin/HIV data.

A number of other people contributed to this book, directly or indirectly. Besides sharing some data, Ray Carroll has always been willing over the years to answer questions and he also looked over a portion of Chapter 10. My measurement error education owes a lot to reading the works of, listening to many talks by, and having conversations with Ray, Len Stefanski, and David Ruppert. After years of a lot of traveling to collaborate with others, I certainly appreciate having John Staudenmayer, and his original thinking, here at the University of Massachusetts. Our joint work forms the basis of much of Chapter 12. I can't thank Meng-Shiou Shieh enough for her reading of the manuscript, double checking many of the calculations, picking up notational inconsistencies and helping me clean up the bibliography. Magne Thoresen generously provided comments on the first draft of Chapter 7 and Donna Spiegelman advised on the use of validation data. I am grateful to Graham Dunn for his reviews of Chapters 2–5 and Gene Buonaccorsi for his assistance with creating the index. Of course, none of those mentioned are in any way responsible for remaining shortcomings or errors. I want to thank Hari Iyer for being my adviser, mentor, collaborator, and friend over the years. On the lighter side, the Coffee Cake Club helped keep me sane (self report) over the years with the Sunday long runs.

Over the years I have been fortunate to receive support for some of my measurement error work from a number of sources. In addition to EPA-SIMS, mentioned earlier, this includes the National Cancer Institute, the U.S. Department of Agriculture and the Division of Mathematical Sciences at the NSF. The University of Massachusetts, in general, and the Department of Mathematics and Statistics in particular, has provided continuing support over the years, including in the computational area. I especially appreciate the assistance of the staff in the departmental RCF (research computing facility). Finally, I am extremely grateful to the Norwegian Research Council for supporting my many visits to collaborate with my colleagues in the Medical School at the University of Oslo.

I appreciate the guidance and patience of Rob Calver of Taylor & Francis throughout this project as well as the production assistance from Karen Simon and Marsha Pronin. I am grateful to John Wiley and Sons, the American Statistical Association, the Biometrics Society, Blackwell Publishing, Oxford University Press, Elsevier Publishing and Taylor & Francis for permissions to use tables and figures from earlier publications. Finally, I want to thank my wife, Elaine Puleo, for her constant support, editing Chapter 1 and serving as a sounding board for my statistical ideas throughout the process of writing of the book.

Computing

Computing was occasionally an obstacle. While there are a number of programs, described below, which handle certain models and types of measurement errors, there were many techniques that I wanted to implement in the examples for which programs were not available. While I had built up some SAS programs to handle measurement error over the years, many of the examples in the book required new, and extensive, programming. There are still examples where I would have liked to have illustrated more methods, but I needed to stop somewhere.

The software used in the book, and some associated programs, include:

- STATA routines rcal (regression calibration) and simex. These, along with the related qvf routine, are available through `http://www.stata.com/merror/`.
- STATA gllamm routines based on Skrondal and Rabe-Hesketh (2004). Available at `http://www.gllamm.org/`. This includes the cme command for parametric maximum likelihood used in a couple of places in the book, as well as some other measurement error related programs.
- SAS macro Blinplus from Spiegelman and colleagues at Harvard University. Available at `http://www.hsph.harvard.edu/faculty/spiegelman/blinplus.html`. This corrects for measurement error using external and internal validation data, based on a linear Berkson model.
- SAS macro from Spiegelman and colleagues at Harvard University which correct for measurement error under additive error with the use of replicate values. This is available at `http://www.hsph.harvard.edu/faculty/spiegelman/relibpls8.html`. This was not used in the book, since the rcal command in STATA was used instead.

The remainder of the computing was carried out using my own SAS programs. Any of these available for public use can be found at
`http://www.math.umass.edu/~johnpb/meprog.html`.

Although not spelled out in any real detail in the book (except for a few comments here and there), there are other programs within the standard software packages that will handle certain analyses involving measurement errors. This includes programs that treat structural equation or latent variables models, instrumental variables and mixed models.

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List of Examples

Other applications not in worked examples (in order of appearance)